

Functional Annotation Report: PP_0860 (UniProt Q88PJ0) of *Pseudomonas putida* KT2440

Organism: *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950 / KT2440) — proteome PSEPK **Gene / locus:** PP_0860 (ordered locus name); EMBL AAN66485.1 **Protein length / mass:** 849 aa / 93.7 kDa **Database annotation (automated):** "Sulfite reductase, flavoprotein component"; KEGG K00380; EC 1.8.1.2

Summary

The gene **PP_0860** (UniProt **Q88PJ0**) of *Pseudomonas putida* KT2440 encodes an **849-amino-acid (~93.7 kDa) integral inner-membrane, iron-regulated diflavin NADPH oxidoreductase**. Its C-terminal cytoplasmic module is a **CysJ / NADPH-cytochrome-P450-reductase-type diflavin reductase** that binds one **FAD** and one **FMN** and catalyzes electron transfer from the two-electron donor **NADPH** to a one-electron acceptor via the canonical relay **NADPH → FAD → FMN → acceptor**. This catalytic module is fused to an N-terminal **polytopic PepSY-associated transmembrane domain (~4 predicted TM helices)** that anchors the protein in the cytoplasmic (inner) membrane.

Although UniProt/EMBL and KEGG label the protein a "sulfite reductase flavoprotein" (EC 1.8.1.2 / K00380), the investigation concludes that this is an **automated homology transfer** derived from the reductase module's sequence similarity to CysJ, not a validated functional assignment. The protein is markedly larger than a canonical soluble *E. coli* CysJ (~600 aa), carries a membrane anchor that CysJ lacks, belongs to the PANTHER "iron-regulated inner-membrane protein" family, and lies immediately adjacent to a TonB-dependent siderophore receptor. The dedicated sulfite-reductase hemoprotein partner (*cysI*) is encoded at a separate, unlinked locus, and a second paralog also carries the K00380 tag — hallmarks of homology-driven annotation rather than genuine pathway membership.

Integrating domain architecture, membrane topology, protein-family clustering, and genomic synteny, the most parsimonious model is that **PP_0860 is a membrane-associated iron-related redox enzyme — most plausibly a ferric-siderophore/partner reductase — that uses NADPH-derived reducing equivalents to reduce an iron-containing acceptor at the cytoplasmic membrane, contributing to iron acquisition/utilization rather than bulk assimilatory sulfite reduction.** The domain architecture, cofactor identity, and localization are high-confidence bioinformatic conclusions; the precise physiological electron acceptor is an evidence-based inference that has **not been experimentally validated** for this specific protein.

Key Findings

Finding 1 — PP_0860 is a membrane-anchored diflavin (FMN + FAD) NADPH oxidoreductase, not a canonical soluble CysJ

Analysis of the 849-residue sequence (UniProt, InterPro, Pfam, CDD, PANTHER, CATH) reveals a **two-module fusion protein**. The **N-terminal module (~residues 1–370)** is a *PepSY-associated transmembrane region* (InterPro **IPR005625** / Pfam **PF03929**; eggNOG **COG3182**, "iron-regulated membrane protein"). The **C-terminal module** is a diflavin reductase comprising a flavodoxin-like FMN-binding domain (IPR001094 / PF00258) and a ferredoxin-NADP-reductase-like FAD/NAD(P)H-binding domain (IPR017927, IPR001709 / PF00175).

A Kyte–Doolittle hydropathy analysis (window = 19) identified **four hydrophobic, membrane-spanning segments** within residues ~1–370, with hydropathy peaks of 1.9–2.0 at approximately aa 5–35, 139–162, 184–213 and 342–369 — the signature of a **polytopic N-terminal inner-membrane anchor** that is entirely absent from bona fide cytoplasmic CysJ. The conserved diflavin catalytic motifs are present in the C-terminal module in the expected order:

Motif	Approx. residue	Function
Flavodoxin FMN loop ASQSGFAEQ	~398–406	FMN cofactor binding
FAD/isoalloxazine motif GSGWLTE	~683	FAD binding
FNR pyrophosphate/NAD(P) motif GNGTGLAGL	~719–727	NAD(P)H binding

At **849 aa / 93.7 kDa**, PP_0860 is substantially larger than the ~600-aa soluble *E. coli* CysJ, the extra mass being accounted for by the N-terminal membrane domain. The joint presence of FMN- and FAD-binding modules plus an NADPH-binding motif places PP_0860 firmly in the **diflavin reductase superfamily**, whose members "tightly bind two flavin cofactors, FAD and FMN, and catalyze transfer of the reducing equivalents from the two-electron donor NADPH to a variety of one-electron acceptors" (PMID: 15063311). The canonical mechanism — "FAD accepts a hydride ion from NADPH, and reduced FAD donates electrons to FMN, which in turn transfers electrons to the heme center of cytochrome P450 or NOS oxygenase domain" (PMID: 22982532) — is the electron relay that PP_0860's C-terminal module is expected to perform.

Finding 2 — Genomic context and paralogy indicate the "sulfite reductase flavoprotein" annotation is automated homology transfer

The KEGG orthology group **K00380** (sulfite reductase NADPH flavoprotein alpha subunit, *cysJ*; EC 1.8.1.2) is assigned to **two** loci in *P. putida* KT2440: **PP_0860** and **PP_1703**. Critically, the cognate sulfite-reductase hemoprotein beta subunit *cysI* (K00381) is encoded at a **separate, unlinked locus, PP_2371**. A functional assimilatory sulfite reductase requires the physical partnership of the CysJ flavoprotein with the CysI hemoprotein; PP_0860 is not co-localized with any *cysI* partner.

The immediate genomic neighborhood of PP_0860 is dominated by **iron-acquisition functions**, not sulfur assimilation:

Locus	Annotation
PP_0858 / PP_0859	Amino-acid metabolism
PP_0860	Target — membrane diflavin oxidoreductase
PP_0861	TonB-dependent outer-membrane catecholate siderophore receptor (K16090)
PP_0862	PKHD-type Fe(II)/2-oxoglutarate-dependent hydroxylase

By contrast, the true CysJ-like paralog PP_1703 is flanked by unrelated genes (a leucine-rich-repeat protein and a hypothetical protein), and the CysI hemoprotein PP_2371 is not in a sulfur-assimilation operon with either flavoprotein. The N-terminal membrane PepSY domain (COG3182, "iron-regulated membrane protein") of PP_0860 is absent from genuine cytoplasmic CysJ proteins. Together, the membrane topology, the iron-themed genomic context, and the absence of a co-localized hemoprotein partner argue strongly that the "sulfite reductase flavoprotein" label is an artifact of automated annotation propagated from the diflavin

domain's homology to CysJ. Notably, sulfite-reductase-type diflavin flavoproteins possess a promiscuous reductase side-activity: "the flavin reductase activity may function during activation of ribonucleotide reductase or during ferrisiderophore reduction" (PMID: 7657631) — an activity directly relevant to PP_0860's iron-acquisition neighborhood.

Finding 3 — PP_0860 belongs to the PANTHER "iron-regulated inner-membrane protein" family; its reductase module is CysJ-like (SiR_like1)

The InterPro entry set for Q88PJ0 includes the PANTHER family **PTHR34219 "IRON-REGULATED INNER MEMBRANE PROTEIN-RELATED"** together with the InterPro family **IPR005625 "PepSY-associated TM protein"** (Pfam PF03929). The reductase module maps to CDD **cd06200 "SiR_like1"** (sulfite-reductase-flavoprotein-like) and to CATH superfamilies **3.40.50.360** (flavodoxin/FMN-binding) and **3.40.50.80** (FNR nucleotide-binding), and it carries the PRINTS signatures **FLAVODOXIN (PR00369)** and **FPNCR (PR00371)**.

This composite classification confirms two things simultaneously: (i) PP_0860 is placed by family clustering into an *iron-regulated inner-membrane* group, and (ii) its catalytic module is of the CysJ/sulfite-reductase-flavoprotein structural type. These are not contradictory — the SiR-type diflavin fold is a versatile electron-transfer scaffold that appears in "naturally occurring flavocytochrome fusion enzymes like nitric oxide synthases (NOS) and the fatty acid hydroxylase from *Bacillus megaterium*" (PMID: 15063311); i.e., the same fold is repeatedly recruited into fusion proteins with distinct physiological roles.

The comparison with the other K00380 locus, PP_1703, further undermines any textbook-CysJ interpretation of either *P. putida* locus. PP_1703 is a very different, 1341-aa **cytoplasmic** protein (0 predicted TM helices) whose N-terminus (MANSEVRSVCPYCGVGCGIV...) carries a **CxxCxxx C [4Fe-4S] / Mo-bis-MGD cysteine motif** characteristic of assimilatory nitrate/nitrite reductase catalytic subunits — a molybdoenzyme. Thus *neither* K00380-annotated locus in KT2440 is a conventional ~600-aa soluble CysJ: PP_0860 is a membrane-anchored diflavin fusion, and PP_1703 is a large cytoplasmic molybdoenzyme.

Finding 4 — Independent literature precedent supports an iron/ferric-siderophore reductase role

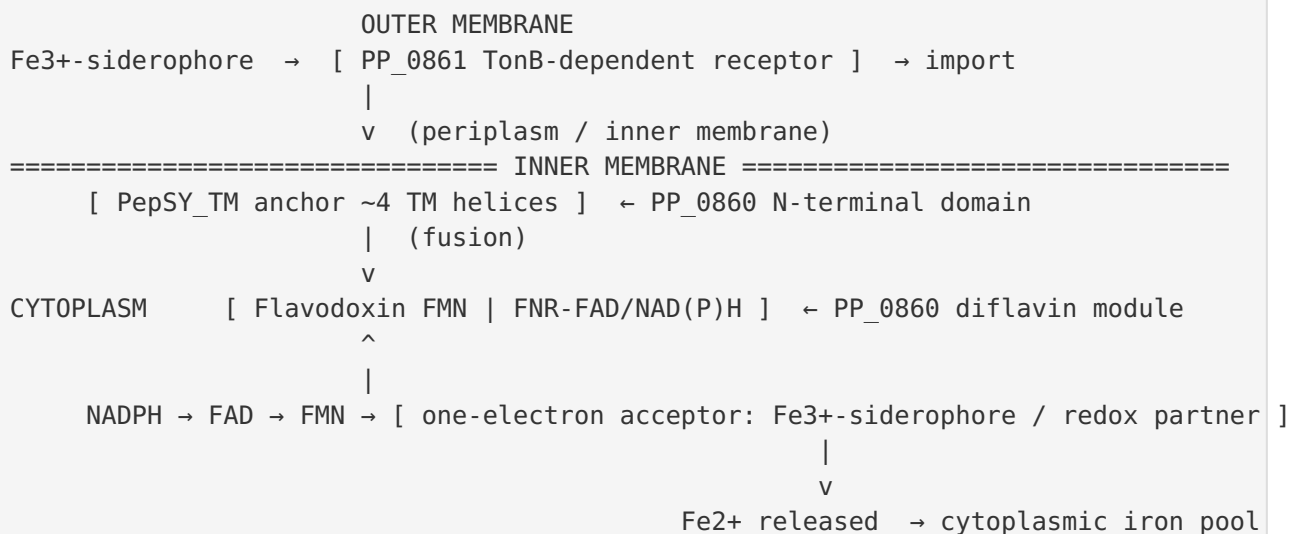
Multiple experimentally characterized bacterial systems establish the physiological niche inferred for PP_0860 — reduction of Fe^{3+} to release iron from internalized siderophores. Three precedents are especially relevant:

1. **Integral inner-membrane siderophore reductases.** A structurally defined "novel family of integral membrane siderophore reductases" in Gram-negative bacteria has been described ([PMID: 34417315](#)), providing direct precedent for an integral-membrane protein performing ferric-siderophore reduction — the topology PP_0860 exhibits.
2. **Dedicated reductases are obligatory for iron release.** Ferric-siderophore reductases (FSR, e.g. FhuF) and siderophore-interacting proteins (SIP) are required because "once inside the cell, iron removal does not occur spontaneously, instead this process is mediated by siderophore-interacting proteins (SIP) and/or by ferric-siderophore reductases (FSR)" ([PMID: 33559753](#)).
3. **Membrane/periplasm-associated ferric-siderophore reduction in *Pseudomonas*.** In the closely related *P. aeruginosa* pyoverdine pathway, ferric-siderophore reduction near the periplasm/membrane releases iron from internalized PVDI-Fe ([PMID: 31924850](#)).

PP_0860 unites the salient features of these systems: it is an integral inner-membrane, iron-regulated protein (PANTHER PTHR34219), physically adjacent to a TonB-dependent catecholate-siderophore receptor (PP_0861), and its diflavin module has the documented flavin/ferrisiderophore-reductase capability of SiR-flavoproteins ([PMID: 7657631](#)). PP_0860 would, however, use a **distinct NADPH→FAD→FMN diflavin mechanism** rather than the fold of the FpvG-type integral-membrane reductase family — i.e., it is a candidate NADPH-dependent membrane iron/ferric-siderophore reductase.

Mechanistic Model / Interpretation

PP_0860 is best described as a **modular membrane oxidoreductase** whose two domains carry out complementary tasks: the N-terminal PepSY_TM domain provides inner-membrane localization (and likely a substrate/partner-interaction surface), while the C-terminal diflavin module performs the electron transfer using cytoplasmic NADPH.



Electron flow. The diflavin module operates by the canonical superfamily mechanism: NADPH delivers a hydride to FAD; reduced FAD passes electrons one at a time to FMN; and FMN delivers single electrons to an external acceptor ([PMID: 15063311](#); [PMID: 22982532](#)). In canonical diflavin enzymes the terminal acceptor is a heme (cytochrome P450, NOS) or the CysI siroheme. In PP_0860, the genomic and family context points instead to an **iron-containing acceptor** — a ferric-siderophore complex or intermediary redox partner — with reduction of Fe³⁺ to Fe²⁺ weakening iron-siderophore affinity and releasing iron into the cytoplasm.

Why not sulfite reduction? A functional assimilatory sulfite reductase (EC 1.8.1.2) is an $\alpha_8\beta_4$ hemoflavoprotein requiring tight association of the CysJ flavoprotein with the CysI siroheme-[4Fe-4S] hemoprotein. In KT2440, CysI (PP_2371) is genetically unlinked to PP_0860, and PP_0860 additionally carries a membrane anchor incompatible with the soluble cytoplasmic CysJ role. The "sulfite reductase flavoprotein" annotation is therefore best read as "**has a CysJ-like diflavin fold**" rather than a validated catalytic assignment.

Localization. All evidence converges on the **cytoplasmic (inner) membrane**: four predicted TM helices, the PepSY-associated TM domain, and PANTHER classification as an "iron-regulated inner-membrane protein." The catalytic diflavin module faces the **cytoplasm**, where it can access NADPH, while the membrane anchor positions the enzyme to interact with imported iron-siderophore substrates or membrane redox partners.

Confidence. This model is built on domain architecture, hydropathy, protein-family clustering, genomic synteny, and analogy to characterized systems. It is well-supported bioinformatically but **not yet experimentally validated** for PP_0860 specifically.

Evidence Base

PMID	Title (abbreviated)	Role in this report
15063311	<i>Electron transfer by diflavin reductases</i>	Defines the diflavin superfamily (FAD+FMN, NADPH→one-electron acceptor); notes SiR-flavoprotein folds occur in fusion flavoenzymes. Supports Findings 1 & 3.
22982532	<i>NADPH-cytochrome P450 oxidoreductase: prototypic diflavin reductase</i>	Describes the NADPH→FAD→FMN→acceptor mechanism PP_0860's module is expected to use. Supports Finding 1.
7657631	<i>Flavin reductase activity of the flavoprotein component of E. coli sulfite reductase</i>	Documents that SiR-type diflavin flavoproteins possess flavin/ ferrisiderophore reductase side-activity. Supports Findings 2 & 4.
33559753	<i>FhuF, a ferric siderophore reductase from E. coli</i>	Establishes that iron release from internalized siderophores is non-spontaneous and requires dedicated FSR/SIP reductases. Supports Finding 4.
34417315	<i>Integral membrane siderophore reductases (structural insights)</i>	Precedent for an integral inner-membrane siderophore reductase family in Gram-negatives, matching PP_0860's topology. Supports Finding 4.
31924850	<i>Iron acquisition by pyoverdine in P. aeruginosa</i>	Shows membrane/periplasm-associated ferric-siderophore reduction releases iron in a close relative of <i>P. putida</i> . Supports Finding 4.

Contextual literature also reviewed: ferredoxin-NADP⁺ reductase isoform roles ([PMID: 36358515](#)); flavin-dependent N-monooxygenases in siderophore biosynthesis ([PMID: 39155115](#)); hydroxamate xenosiderophore transport ([PMID: 38189440](#)); and the NtrYX two-component system linking iron and denitrification ([PMID: 36012437](#)).

Challenges / caveats. No reviewed paper characterizes PP_0860 directly. The ferric-siderophore-reductase role is an analogy supported by the convergence of topology, iron-regulated family, and genomic synteny — but the characterized reductase precedents (FhuF, the FpvG-type integral-membrane family) are not themselves diflavin PepSY-fusion proteins, so

PP_0860 may be a distinct mechanistic variant. The alternative — that PP_0860 acts as a general NADPH-dependent membrane electron donor to an as-yet-unidentified partner — cannot be excluded.

Supported vs. Refuted Hypotheses

- **Supported (strong bioinformatic support):** PP_0860 is a diflavin (FAD+FMN) NADPH oxidoreductase of the CysJ/CPR/NOS superfamily.
 - **Supported (strong):** PP_0860 is an integral **inner-membrane, iron-regulated** protein via its N-terminal PepSY-associated TM domain.
 - **Refuted / disfavoured:** PP_0860 is the organism's canonical cytoplasmic assimilatory sulfite-reductase α -subunit (CysJ). Its membrane topology, size, iron-regulated family, non-cys genomic context, separate CysI locus, and paralog redundancy argue against this.
 - **Open (leading hypothesis):** PP_0860 functions in **iron acquisition/utilization** as a membrane electron-transfer flavoprotein, possibly a ferric-siderophore reductase or the reductase partner of an iron-handling membrane system.
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Limitations and Knowledge Gaps

1. **No direct experimental data on PP_0860.** Every functional conclusion is an inference from sequence, structure prediction, family membership, and genomic context. No enzyme assays, knockout phenotypes, or localization experiments for PP_0860 itself were found in the literature.
2. **Substrate identity unproven.** The proposed acceptor (ferric-siderophore or membrane redox partner) is inferred from adjacency to PP_0861 and family analogy; the actual physiological electron acceptor and iron-substrate specificity remain unknown.
3. **Membrane topology is predicted, not solved.** The four TM helices come from Kyte–Doolittle hydropathy; precise topology and the disposition of the diflavin module relative to the membrane are experimentally undetermined.
4. **Annotation ambiguity persists in databases.** UniProt/EMBL and KEGG still label PP_0860 a sulfite-reductase flavoprotein. This report argues that label is an automated artifact, but a non-canonical sulfur-redox role cannot be formally excluded without experiments.

5. **Cofactor occupancy unverified.** Although FAD and FMN binding motifs are present, actual cofactor loading and stoichiometry have not been measured for the recombinant protein.

Proposed Follow-up Experiments / Actions

1. **Recombinant expression and cofactor analysis.** Express PP_0860 (full-length and the soluble C-terminal diflavin module); confirm FAD/FMN content by UV-visible spectroscopy and HPLC; measure NADPH-oxidase / cytochrome-c / ferricyanide reductase activity to confirm diflavin electron-transfer function.
 2. **Ferric-siderophore reductase assay.** Test whether the diflavin module reduces Fe³⁺-siderophore complexes (e.g., the catecholate siderophore recognized by PP_0861, or model ferric substrates) with NADPH as electron donor; quantify Fe²⁺ release (ferrozine assay) and kinetic parameters.
 3. **Membrane topology and localization.** Use cell fractionation plus topology mapping (LacZ/PhoA fusions or protease-accessibility) to confirm inner-membrane localization and the cytoplasmic orientation of the catalytic module.
 4. **Genetics and phenotyping.** Construct a PP_0860 deletion mutant in KT2440; assay growth under iron limitation, siderophore-dependent iron uptake, and ⁵⁵Fe accumulation; test genetic interaction with PP_0861.
 5. **Regulation.** Determine whether PP_0860 is Fur-regulated / iron-repressed (qRT-PCR under iron-replete vs. iron-limited conditions; Fur-box search in the promoter) to test the "iron-regulated" family prediction.
 6. **Structural biology.** Pursue a cryo-EM or crystal structure of the full-length membrane protein, or AlphaFold-guided model validation, to define the PepSY_{TM}-diflavin interface and candidate substrate-binding sites.
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Conclusion

PP_0860 (Q88PJ0) of *Pseudomonas putida* KT2440 is an **integral inner-membrane, iron-regulated diflavin (FAD+FMN) NADPH oxidoreductase**, not the canonical cytoplasmic sulfite-reductase flavoprotein its automated annotation states. Its N-terminal PepSY-associated transmembrane domain anchors it in the cytoplasmic membrane while its C-terminal CysJ/

cytochrome-P450-reductase-type module oxidizes NADPH and relays electrons (NADPH → FAD → FMN → acceptor) to a one-electron acceptor facing the cytoplasm. The convergence of membrane topology, iron-regulated protein-family classification, genomic linkage to a TonB-dependent siderophore receptor, and the existence of separate CysJ-like (PP_1703) and CysI hemoprotein (PP_2371) loci points to a **membrane-associated iron-related redox function — most plausibly ferric-siderophore/partner reduction for iron release — rather than bulk assimilatory sulfite reduction**. This is a well-reasoned bioinformatic inference awaiting direct experimental confirmation.

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